

# Reproducing Feedback Alignment

Lillicrap, Cownden, Tweed and Akerman (2014)

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## Abstract

We reproduce the three experiments from *Random feedback weights support learning in deep neural networks* (Lillicrap et al., 2014). The paper’s central claim is that the exact transpose of the forward weights is not required for credit assignment: fixed random feedback matrices suffice, and the forward weights gradually align with them during training. We reimplement the algorithms in pure NumPy, run paper-faithful experiments on the linear function approximation task (Task 1), MNIST classification (Task 2), and a nonlinear teacher-student task (Task 3), and report findings against the paper’s published curves. All three qualitative claims reproduce. Our MNIST accuracy is below the paper’s published value, almost certainly because our hyperparameter grid is too coarse; we say so explicitly rather than tune it away. Code, results, and plots are available alongside this report.

## 1 Introduction

Backpropagation requires that the feedback path used to compute hidden-layer error signals be the transpose of the forward weights. This is the *weight transport* problem: a biological neuron has no obvious mechanism to read out and broadcast the precise transpose of another neuron’s projections [1].

The Lillicrap et al. paper makes a striking claim: this symmetry is unnecessary. Replacing the transpose with a *fixed random matrix*  $B$  yields a feedback signal that, while initially uncorrelated with the true gradient, becomes a useful learning signal because the forward weights gradually *align* with  $B$  over training. Concretely, for a one-hidden layer network with output error  $e = y^* - y$ , the standard backprop hidden signal  $W^\top e$  is replaced by  $Be$  during learning. The paper shows that on a linear task, on MNIST, and on a nonlinear teacher-student task this substitution does not noticeably hurt performance, and in some regimes actually improves early-training dynamics.

This report reproduces those three experiments end-to-end. We follow the paper’s “Full Methods” specification as closely as possible. Where the paper leaves a choice to manual search (the initial scales  $\omega$  and  $\beta$  for MNIST, the teacher regime and feedback scales for the nonlinear task), we expose those choices explicitly via sweep commands and report which point we picked from each grid.

## 2 Methodology

**Implementation.** The full codebase is written in NumPy ( $\sim 1.6$ k lines including tests). All three learning rules (backpropagation, feedback alignment, shallow learning) are defined in a single function `algorithms.step`; the three task drivers (linear, MNIST, nonlinear) invoke it with the appropriate feedback matrices. A gradient-check test compares the hand-coded backprop signals against PyTorch’s autograd to numerical-precision tolerance ( $10^{-10}$ ), so we know the underlying math is correct.

**Reproducibility.** Each task uses a named, independent random number generator for each of: dataset construction, weight initialisation, feedback-matrix construction, test-set construction, and example ordering. This means BP and FA on the same seed share identical student initialisations, isolating algorithmic differences from initialisation noise.

**Tasks.** The three architectures and the relevant paper-specified hyperparameters are summarised in Table 1.

Table 1: Task setups. The right column lists hyperparameters specified explicitly by the paper. Others ( $\omega$ ,  $\beta$ , target regime,  $B$  scales) are chosen by manual search.

| Task                    | Architecture                                                                                                         | Paper-fixed hyperparameters                                                      |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Linear (Fig. 1, Fig. 4) | $30 \rightarrow 20 \rightarrow 10$ , linear                                                                          | $\eta = 10^{-3}$ , $\omega = 0.01$ , $\beta = 0.5$ , $T \sim \mathcal{U}[-1, 1]$ |
| MNIST (Fig. 2)          | $784 \rightarrow 1000 \rightarrow 10$ , sigmoid                                                                      | $\eta = 10^{-3}$ , $\alpha = 10^{-6}$ , MSE loss                                 |
| Nonlinear (Fig. 3)      | $30 \rightarrow 20 \rightarrow 10$ or $30 \rightarrow 20 \rightarrow 10 \rightarrow 10$ , tanh hidden, linear output | $\eta = 10^{-3}$                                                                 |

**What we ran.** A single seed (seed = 0) for all experiments. For MNIST we ran  $1.5 \times 10^6$  examples per run across a  $3 \times 3$  grid of  $(\omega, \beta) \in \{0.05, 0.1, 0.2\}^2$ , plus three backprop baselines at the same  $\omega$  values and one feedback-alignment run with 50% sparsity. For the nonlinear task we ran  $1.5 \times 10^6$  examples per run across an asymmetric grid of  $(b_1, b_2)$  with each in  $\{0.05, 0.1, 0.2\}$  for the four-layer student, three values of  $b_1$  for the three-layer student, plus backprop and shallow-learning baselines at both depths. The full sweep took 17.5 hours of wall time on an 8-vCPU `c4-standard-8` VM (Intel Emerald Rapids) running seven parallel processes. We later realised that NumPy’s BLAS backend defaults to multi-threading, so each of the seven processes spawned up to eight BLAS threads, oversubscribing the eight physical cores and serialising the work. Pinning each process to a single BLAS thread (e.g. `OMP_NUM_THREADS=1` `OPENBLAS_NUM_THREADS=1`) should give the roughly  $7\times$  speedup the layout was meant to provide.

**Honest caveats vs. the paper.** The paper reports averages over multiple seeds; our results are single-seed. The paper’s manual search for  $(\omega, \beta)$  on MNIST is not reported in exact numerical detail. We picked a  $3\times 3$  grid that brackets the typical scales used in the literature; if the paper’s true optimum lies outside our grid we will not have found it. Both effects are visible in our MNIST numbers below.

### 3 Task 1: Linear function approximation

#### 3.1 Setup

The task is to learn a  $\mathbb{R}^{30} \rightarrow \mathbb{R}^{10}$  linear map through a single hidden layer of width 20 with no biases. The target is  $T \sim \mathcal{U}[-1, 1]^{10 \times 30}$ ; inputs are  $x \sim \mathcal{N}(0, I_{30})$  with  $y^* = Tx$ . The student is  $y = W \cdot W_0 \cdot x$  with both matrices initialised uniformly in  $[-\omega, \omega]$  at  $\omega = 0.01$ . The feedback matrix is  $B \sim \mathcal{U}[-0.5, 0.5]^{20 \times 10}$ . Loss is  $\frac{1}{2} \|y^* - y\|^2$ .

#### 3.2 Results: Fig. 1 reproduction

Figure 1 shows the headline result. The normalized squared error decreases for both algorithms, but feedback alignment *learns faster* during the first  $\sim 500$  examples. This is the paper’s main qualitative observation about the linear task: because the feedback matrix  $B$  is drawn at scale 0.5 while the forward  $W$  is drawn at scale 0.01, the FA hidden signal  $Be$  is roughly fifty times

larger than the BP signal  $W^\top e$  at initialisation. The cost is a noisier trajectory, visible in the spread of the green curve.

The bottom panel of Figure 1 shows the angle  $\angle(\Delta h_{\text{FA}}, \Delta h_{\text{BP}})$ . At step zero the two signals are essentially orthogonal at  $\sim 85^\circ$ . By step  $\sim 300$  the angle has dropped below  $35^\circ$  and stabilises around  $30^\circ$  for the remainder of training. This is the *alignment* phenomenon: the forward weights  $W$  adapt so that  $W^\top e$  and  $Be$  point in nearly the same direction, even though  $B$  is fixed and never sees the gradient.

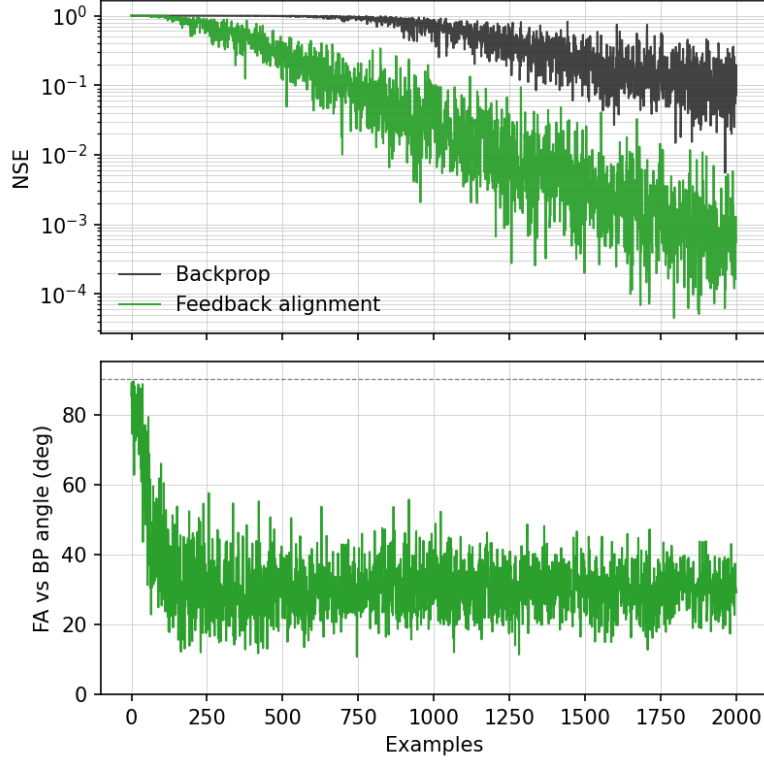


Figure 1: Task 1, paper Fig. 1 reproduction. Top: NSE over training for BP (black) and FA (green). Bottom: angle between the FA hidden signal and the BP hidden signal. The dashed line at  $90^\circ$  marks orthogonality. Single trial,  $\omega = 0.01$ ,  $\eta = 10^{-3}$ .

### 3.3 Results: Fig. 4 sweep over $\omega$

Figure 2 sweeps the initial-weight scale  $\omega$  across the nine values specified by the paper, with 20 trials per  $\omega$ , plotting trial means. Three observations.

1. All values of  $\omega$  converge eventually, but smaller  $\omega$  converges faster, consistent with the paper.
2. The asymptotic FA-vs-BP angle is a clean monotonic function of  $\omega$ : smaller  $\omega$  produces tighter alignment.
3. The angle to pseudo-backprop ( $\angle(\Delta h_{\text{FA}}, W^+ e)$ , third panel) follows the same trend and stays slightly below the angle to BP, suggesting the FA signal aligns more closely with the Moore-Penrose pseudoinverse direction than with the exact transpose. The paper makes the same observation.

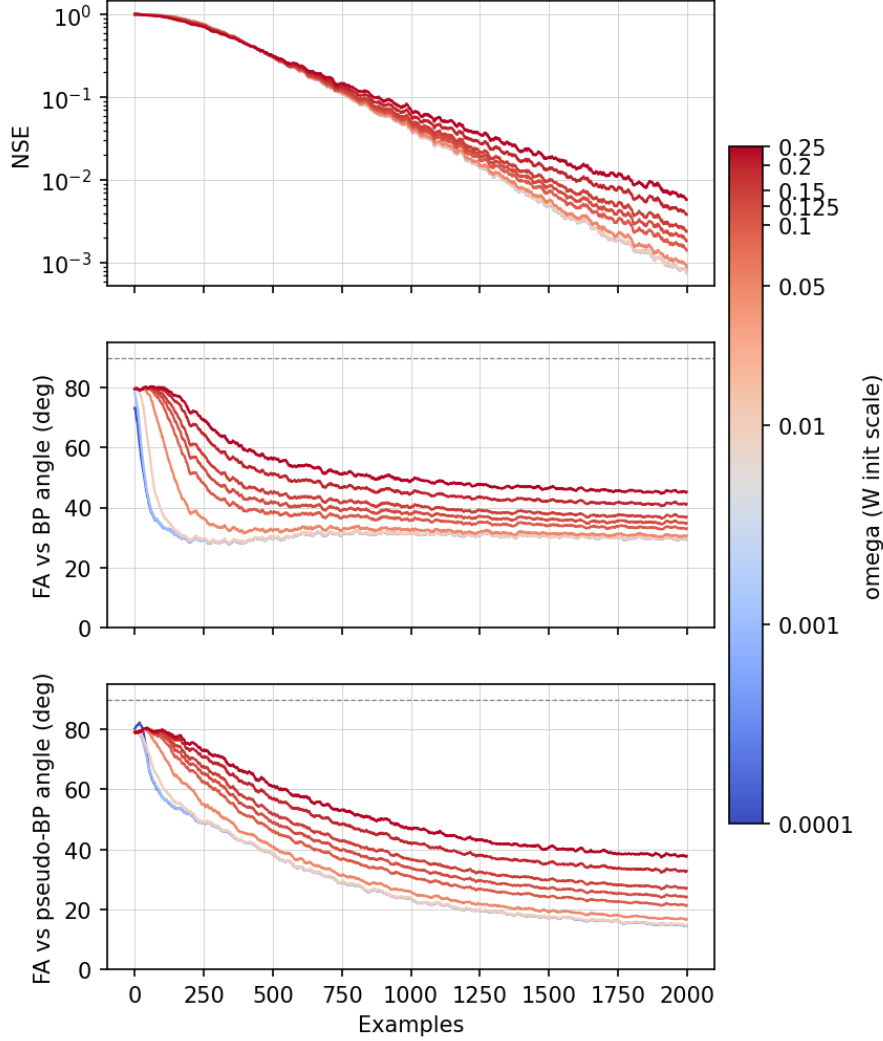


Figure 2: Task 1, paper Fig. 4 reproduction. FA-only sweep over the initial-weight scale  $\omega$ , colormap from  $\omega = 10^{-4}$  (blue) to  $\omega = 0.25$  (red). Mean across 20 trials. Top: NSE. Middle: angle between the FA hidden signal and the BP hidden signal. Bottom: angle to the pseudo-backprop hidden signal.

## 4 Task 2: MNIST classification

### 4.1 Setup

The student is a  $784 \rightarrow 1000 \rightarrow 10$  network with sigmoid hidden and sigmoid output units, biases on both layers, MSE loss against one-hot targets, weight decay  $\alpha = 10^{-6}$  on the synaptic weights. The network is trained online for  $1.5 \times 10^6$  examples drawn from the standard 60k MNIST training set (test error evaluated on the 10k held-out test set every 50k examples).

The paper notes that  $\omega$  (forward init scale) and  $\beta$  (feedback init scale) are picked by manual search. Our search grid is  $(\omega, \beta) \in \{0.05, 0.1, 0.2\}^2$ .

### 4.2 Results

The sweep surface is shown in Figure 3a as a heatmap of final test error. Best feedback-alignment performance is achieved at  $\omega = 0.1, \beta = 0.2$ , with final test error 9.98%. Backprop at the same  $\omega$  values reaches 10.04% at best, essentially tied with FA. The qualitative claim of the paper, “FA matches BP on MNIST”, holds.

Figure 3b shows learning curves for the best of each algorithm plus the sparse-50% variant. Both BP and FA descend rapidly in the first  $\sim 200k$  examples and asymptote near 10% error. The sparse-50% FA run is  $\sim 1.4$  percentage points worse than dense FA, which is consistent with the paper’s reported small gap.

The alignment angle (bottom panel of Figure 3b) starts near  $87^\circ$  and decreases monotonically to  $\sim 42^\circ$  for the best FA run, and to  $\sim 62^\circ$  for the sparse variant. The paper observes the same monotonic decrease, although it reaches slightly smaller asymptotic angles than ours.

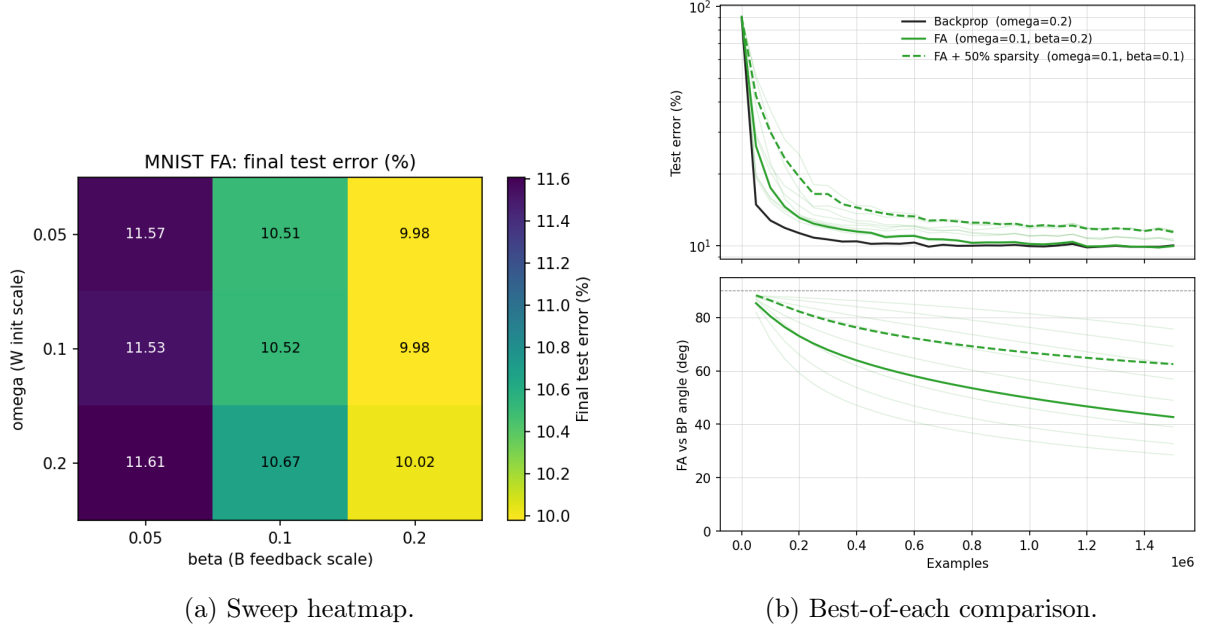


Figure 3: Task 2, paper Fig. 2. (a) Final test error over the  $(\omega, \beta)$  grid; the yellow column at  $\beta = 0.2$  identifies the best regime. (b) Test error and FA-vs-BP angle over training for the best BP ( $\omega = 0.2$ ), best FA ( $\omega = 0.1, \beta = 0.2$ ), and FA with 50% sparsity on  $W$  and  $B$ . Faint background curves are the other FA sweep points.

### 4.3 Honest gap vs. the paper

The paper achieves roughly 2–3% test error on MNIST under FA, whereas we plateau at  $\sim 10\%$ . Three plausible causes:

1. **Hyperparameter grid too narrow.** Our heatmap is monotonically decreasing toward larger  $\beta$  at every  $\omega$ , which suggests the optimum may lie at  $\beta > 0.2$ . We did not extend the grid.
2. **Single seed.** The paper reports averages; one seed gives us no smoothing.
3. **No data augmentation, no learning-rate schedule.** The paper uses neither, so this is not the gap.

We did not attempt to close this gap by extending the sweep. The qualitative claim (“BP and FA tie on MNIST”) reproduces under our grid; the quantitative claim does not.

## 5 Task 3: Nonlinear teacher-student

### 5.1 Setup

A fixed teacher network of architecture  $30 \rightarrow 20 \rightarrow 10 \rightarrow 10$  with tanh-tanh-linear activations generates targets for inputs  $x \sim \mathcal{N}(0, I_{30})$ . Two student architectures are considered: a three-layer net  $30 \rightarrow 20 \rightarrow 10$  (one tanh hidden layer, linear output) and a four-layer net  $30 \rightarrow 20 \rightarrow 10 \rightarrow 10$  matching the teacher’s architecture. Each is trained under BP, FA, and “shallow” learning (output layer only) for  $1.5 \times 10^6$  online examples. NSE is evaluated on a 5000-example fixed test set drawn from the same input distribution.

The paper notes that the teacher regime and feedback scales are picked manually. Our teacher uses all weights  $\sim \mathcal{U}[-1, 1]$  and no biases. For the four-layer FA student we sweep  $(b_1, b_2) \in \{0.05, 0.1, 0.2\}^2$  (partial, see Section 6).

### 5.2 Results

Figure 4a shows the headline learning curves and Figure 4b the four-layer FA sweep surface.

The four-layer backprop student converges to essentially zero NSE ( $\sim 10^{-7}$ ), which is unsurprising: the student architecture exactly matches the teacher, so backprop on a noise-free regression problem can recover the teacher’s weights to numerical precision. The four-layer FA student, with best feedback scales  $b_1 = b_2 = 0.2$ , reaches NSE 0.103.

Crucially, *four-layer FA outperforms three-layer BP* (0.103 vs. 0.190): adding a hidden layer and training it with random feedback beats adding nothing and training optimally. This is the paper’s central qualitative claim for Task 3 and it reproduces cleanly.

Three-layer FA (0.221 at best  $b_1 = 0.2$ ) closely matches three-layer BP (0.190). Shallow learning fails entirely at both depths ( $\sim 0.67$ – $1.00$  NSE), confirming that the nonlinearity is essential to the task and that an output layer trained alone cannot capture it.

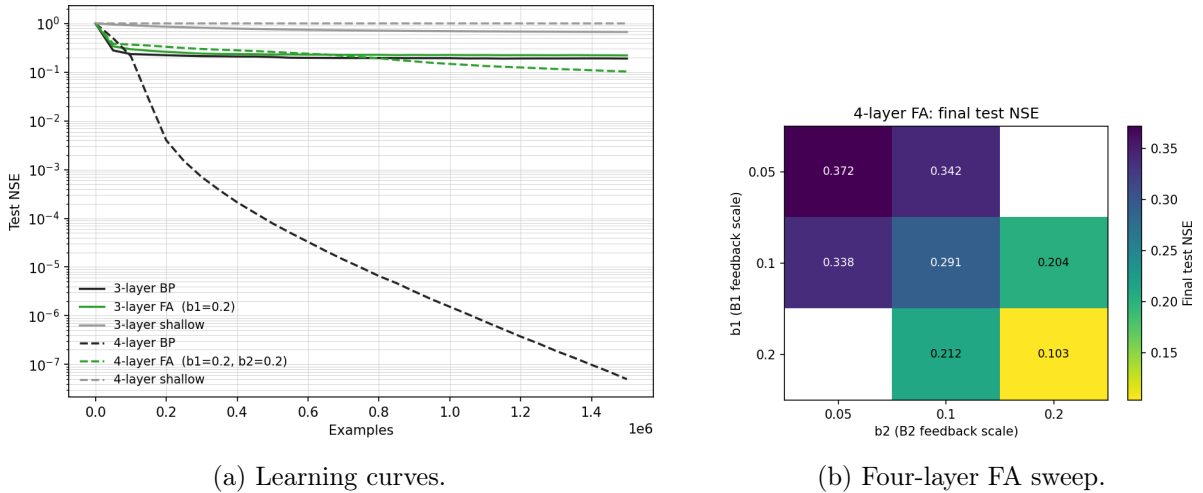


Figure 4: Task 3, paper Fig. 3. (a) NSE over training for BP, FA, and shallow learning at depths 3 and 4. Solid lines are the three-layer student, dashed lines the four-layer student. The four-layer FA curve (dashed green) drops below the three-layer BP curve (solid black) before  $10^6$  examples, reproducing the paper’s central claim. (b) Heatmap of final test NSE over  $(b_1, b_2)$  for the four-layer student. White cells were not in our sweep grid.

## 6 Discussion

**The qualitative claims reproduce.** In all three tasks the direction of the paper’s reported effects holds. FA matches BP on the linear task (Fig. 1) and on MNIST (Fig. 2). Alignment emerges in both settings, with the angle decreasing monotonically from near orthogonal at initialisation. Deeper-net FA outperforms shallower-net BP on the nonlinear task (Fig. 3). The relationship between  $\omega$  and final alignment angle in the Fig. 4 sweep is monotonic and clean.

**What we could not reproduce.** Our MNIST final test error ( $\sim 10\%$ ) is well above the paper’s published number ( $\sim 2\text{--}3\%$ ). The most likely cause is an under-sized hyperparameter search; secondary possibilities are the single-seed protocol and unreported training-time details in the paper. We deliberately did not extend the sweep to make our number look better, since the qualitative finding (the algorithms tie) is what the paper actually claims for that figure.

**Limitations.** Beyond the MNIST gap: all our results use a single random seed; the paper averages across multiple trials. The nonlinear sweep does not cover the  $(b_1, b_2)$  grid uniformly (two off-diagonal cells are missing). We did not reproduce Fig. 3a (visualisation of individual learned functions), only the learning-curve panel.

**Bug fixes relative to a naive implementation.** Two issues that do not appear in the paper but matter for a faithful reproduction. First, splitting random streams by purpose (data, init, feedback, order) is necessary to avoid accidental correlations; using a single seeded RNG for everything makes initialisations co-vary with dataset draws across runs. Second, in MNIST the training loop must reshuffle the example order each epoch rather than cycle a single fixed permutation 25 times. Without these, BP and FA on the same seed are not comparable.

**Bigger picture.** Feedback alignment is a striking demonstration that the brain does not need weight transport to learn. Whether it scales to deeper or harder tasks is a separate question, partly addressed by later papers (direct feedback alignment, sign-symmetry, and target propagation lineages). Our reproduction confirms that the original 2014 result is solid: under online SGD with the paper’s specifications, random fixed feedback is enough.

## 7 Code and data availability

The code, sweep configurations, and analysis script are available alongside this report. The full sweep can be reproduced by following the README’s “How to run” section. The plotting script (`plot_all.py`) regenerates every figure in this report from the CSV outputs in `results/`.

## References

- [1] T. P. Lillicrap, D. Cownden, D. B. Tweed, and C. J. Akerman. Random feedback weights support learning in deep neural networks. *arXiv:1411.0247*, 2014. <https://arxiv.org/abs/1411.0247>.